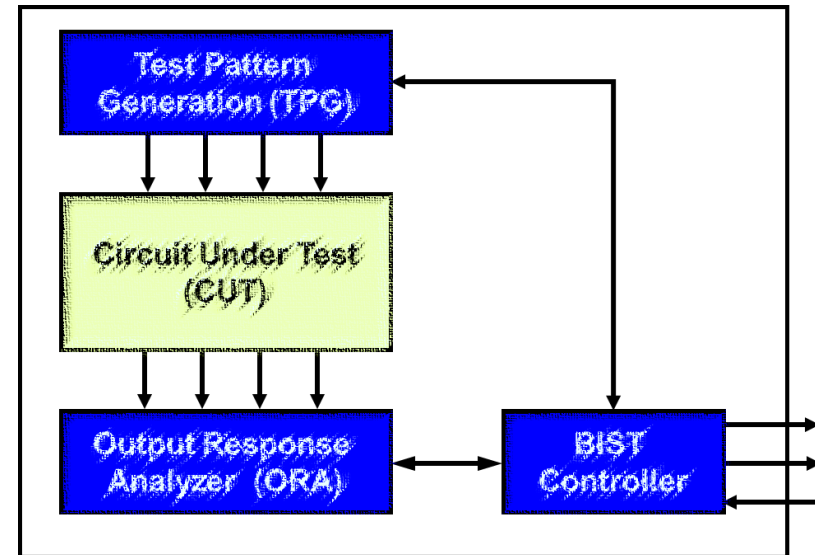


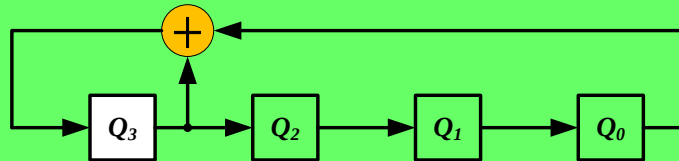
BIST Part1 - TPG

- Introduction
- Test Pattern Generation (TPG)
 - ◆ Deterministic: ROM, Algorithm, Counter
 - ◆ Pseudo Random
 - * Linear Feedback Shift Register, LFSR (1977)
 - Two types of LFSR
 - Design of LFSR
 - ⇒ How to find seed?
 - ⇒ What polynomial?
 - ⇒ All-zero pattern?
 - ⇒ What is LFSR degree?
 - * Cellular Automata, CA (1984)



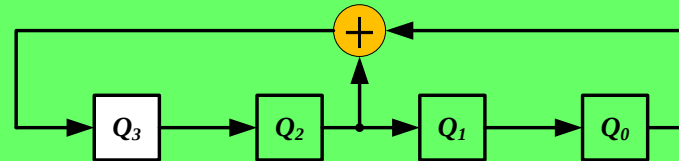
Quiz

Q: Given two LFSR of x^4+x^3+1 , x^4+x^2+1 . seed=1000. Fill in table. $L_c=?$



state	Q_3	Q_2	Q_1	Q_0
0	1	0	0	0
1	1	1	0	0
2	1	1	1	0
3	1	1	1	1
4	0	1	1	1
5	1	0	1	1
6	0	1	0	1
7	1	0	1	0
8	1	1	0	1
9	0	1	1	0
10	0	0	1	1
11	1	0	0	1
12	0	1	0	0
13	0	0	1	0
14	0	0	0	1
15 (=0)	1	0	0	0

$L_c=15$

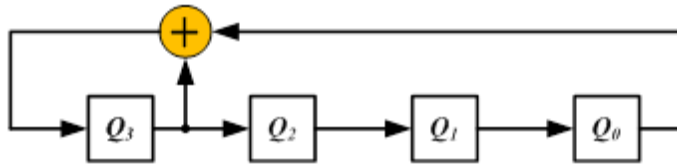


state	Q_3	Q_2	Q_1	Q_0
0	1	0	0	0
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

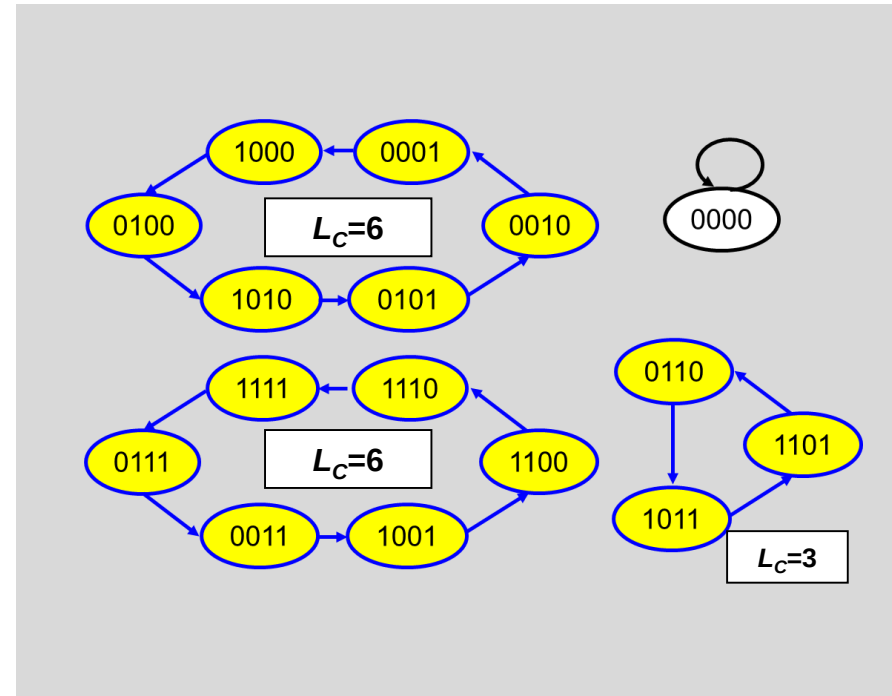
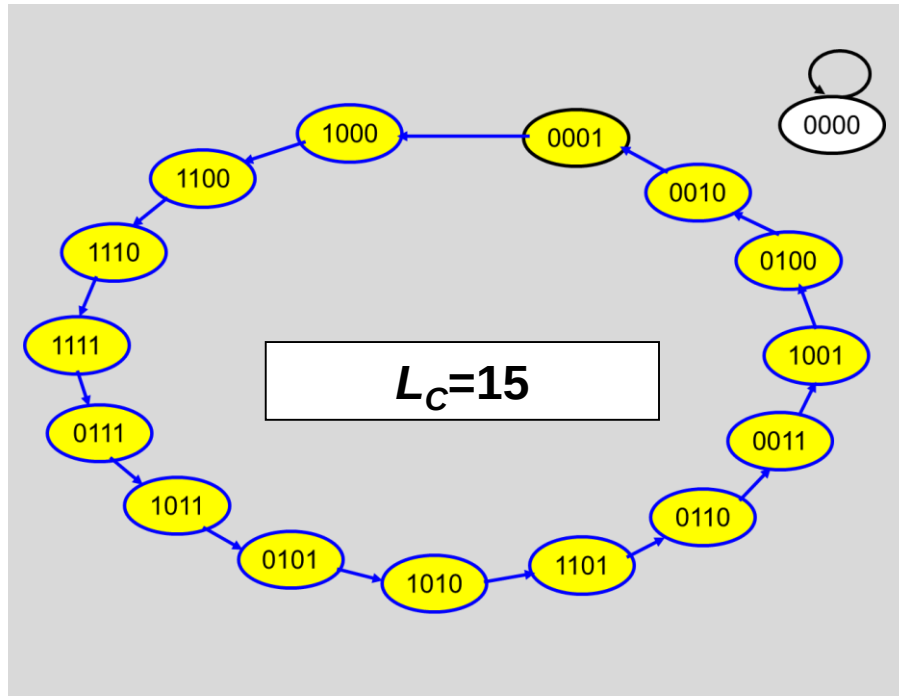
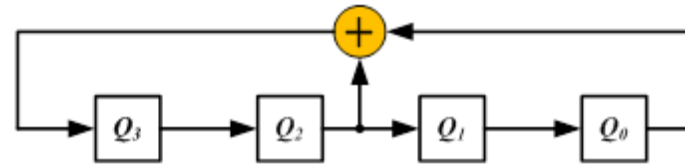
$L_c=?$

Polynomial and L_c

$$x^4 + x^3 + 1$$



$$x^4 + x^2 + 1$$



Larger L_c , Better TPG

Primitive Polynomial

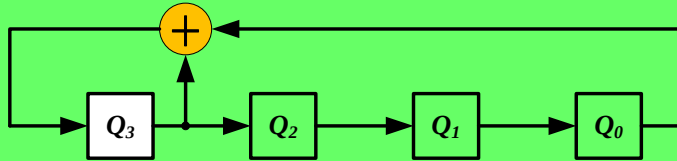
- **Primitive polynomial (PP)** generates *maximum-length sequence*
 - ♦ $L_c = 2^N - 1$, N =degree of PP
- **PP NOT unique for a given N**
 - ♦ $1+x+x^4$, $1+x^3+x^4$ are both PP
- **PP of degree N must satisfy:**
 - ♦ PP is *irreducible*
 - ♦ PP has odd number of terms
 - ♦ PP divides $1+x^{2^N-1}$; $N>3$
 - ♦ more details see (BMS 87)
- PP can be found by **Seive Method**
 - ♦ Many available on line

N	primitive polynomial $f(X)$
1,2,3,4,6,7,15,22	$1 + X + X^N$
5,11,21,29	$1 + X^2 + X^N$
10,17,20,25,28,31	$1 + X^3 + X^N$
9	$1 + X^4 + X^N$
23	$1 + X^5 + X^N$
18	$1 + X^7 + X^N$
8	$1 + X^2 + X^3 + X^4 + X^N$
12	$1 + X + X^2 + X^4 + X^N$
13	$1 + X + X^3 + X^4 + X^N$
14,16	$1 + X^3 + X^4 + X^5 + X^N$
19,27	$1 + X + X^2 + X^5 + X^N$
24	$1 + X + X^2 + X^7 + X^N$
26	$1 + X + X^2 + X^6 + X^N$
30	$1 + X + X^2 + X^{23} + X^N$
32	$1 + X + X^2 + X^{22} + X^N$

PP Often Used to Design TPG

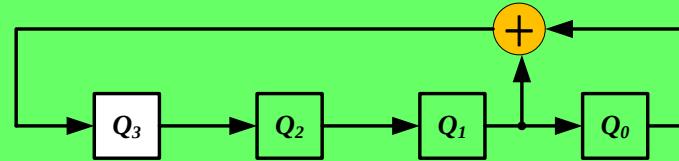
Quiz

Q: Compare two PP x^4+x^3+1 , x^4+x+1 . Fill in table. $L_c=?$



state	Q_3	Q_2	Q_1	Q_0
0	1	0	0	0
1	1	1	0	0
2	1	1	1	0
3	1	1	1	1
4	0	1	1	1
5	1	0	1	1
6	0	1	0	1
7	1	0	1	0
8	1	1	0	1
9	0	1	1	0
10	0	0	1	1
11	1	0	0	1
12	0	1	0	0
13	0	0	1	0
14	0	0	0	1
15 (=0)	1	0	0	0

$L_c=15$

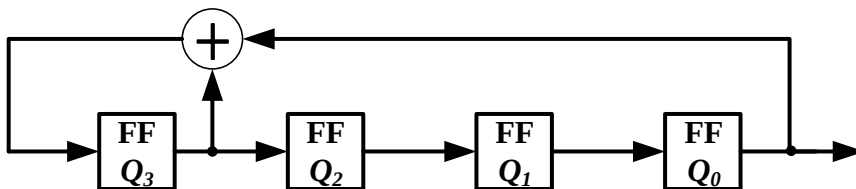


state	Q_3	Q_2	Q_1	Q_0
0	1	0	0	0
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15 (=0)				

$L_c=?$

m-Sequence

- A sequence generated by *primitive polynomial* LFSR is called
 - ♦ **Maximum-length sequence (m-sequence)**
 - ♦ *aka.* **pseudorandom sequence, pseudonoise sequence**
- m-sequence has period **$L_c = 2^N - 1$**
- Example: $N=4$
 - ♦ $x^4 + x^3 + 1$
 - ♦ Period $L_c = 15$



m-sequence=000111101011001
(left bit first)

state	Q ₃	Q ₂	Q ₁	Q ₀
0	1	0	0	0
1	1	1	0	0
2	1	1	1	0
3	1	1	1	1
4	0	1	1	1
5	1	0	1	1
6	0	1	0	1
7	1	0	1	0
8	1	1	0	1
9	0	1	1	0
10	0	0	1	1
11	1	0	0	1
12	0	1	0	0
13	0	0	1	0
14	0	0	0	1

↓ m-sequence

Properties of m-Sequence (1)

- 1. Period $L_c = 2^N - 1$
 - Example: $x^4 + x^3 + 1$, $N=4$
 - 000111101011001
 - $L_c = 15$
- 2. 2^{N-1} ones and $2^{N-1}-1$ zeros (almost equal probability)
 - Example:
 - 8 ones, 7 zeros
- 3. If a window of width N is slid along m-sequence, each of $2^N - 1$ non-zero N -tuple is seen **exactly once** in a period

0	0	0	1	1	1	1	0	1	0	1	1	0	0	1	0	0	0
0	0	0	1														
	0	0	1	1													
		0	1	1	1												
			1	1	1	1											
				1	1	1	0										
					1	1	0	1									
						1	0	1	0								
							0	1	0	1							
								1	0	1	1						
									0	1	1	0					
										1	1	0	0				
											1	0	0	1			
												0	0	1	0		
													0	1	0	0	
														1	0	0	0

Properties of m-Sequence (2)

- 4. m-sequence + its **cyclic shift** = **another cyclic shift** of itself
 - ♦ **cyclic-shift-and-add property**
 - ♦ This property is useful for *phase shifter* (see BIST-2 chapter)
 - ♦ Example: m-sequence of x^4+x^3+1 , $N=4$
 - * NOTE: mod-2 add = XOR

⊕	0	0	0	1	1	1	1	0	1	0	1	1	0	0	1	Original m-sequence
	1	0	0	0	1	1	1	1	0	1	0	1	1	0	0	Right cyclic-shift by 1 bit
	1	0	0	1	0	0	0	1	1	1	1	0	1	0	1	Right cyclic-shift by 4 bits

- See *finite field theory* or (BMS 87) for more properties

m-sequence Has Nice Properties

Quiz

Q: Given another m-sequence **000100110101111** generated by PP x^4+x+1 .

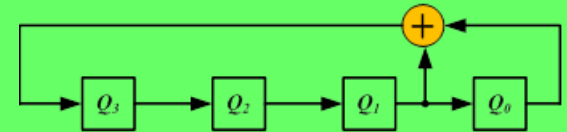
Show that it also satisfies properties:

- 1) $L_c=15$
- 2) 8 ones and 7 zeros
- 3) it has cyclic-shift-add property

original + left-shift-by-2-bit = left-shift-by-?-bit

A:

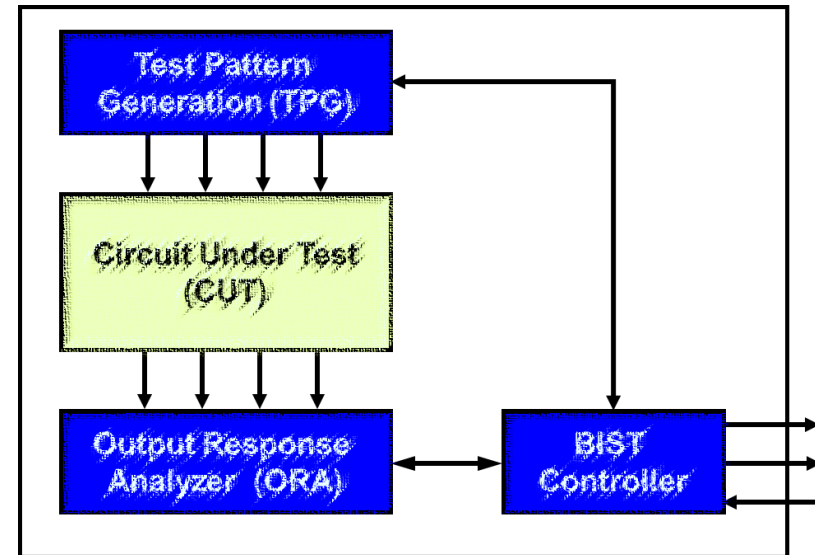
$$\begin{array}{r}
 \oplus \quad \begin{array}{cccccccccccccccc}
 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\
 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0
 \end{array} \\
 \hline
 \begin{array}{cccccccccccccccc}
 \square & \square & \square & \square & \square & \square & \square & \square & \square & \square & \square & \square & \square & \square & \square
 \end{array}
 \end{array}$$



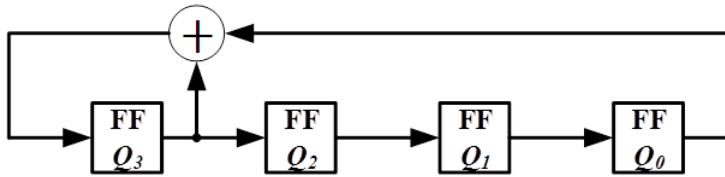
	Q ₃	Q ₂	Q ₁	Q ₀
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	1	0	0	1
4	1	1	0	0
5	0	1	1	0
6	1	0	1	1
7	0	1	0	1
8	1	0	1	0
9	1	1	0	1
10	1	1	1	0
11	1	1	1	1
12	0	1	1	1
13	0	0	1	1
14	0	0	0	1

BIST Part1 - TPG

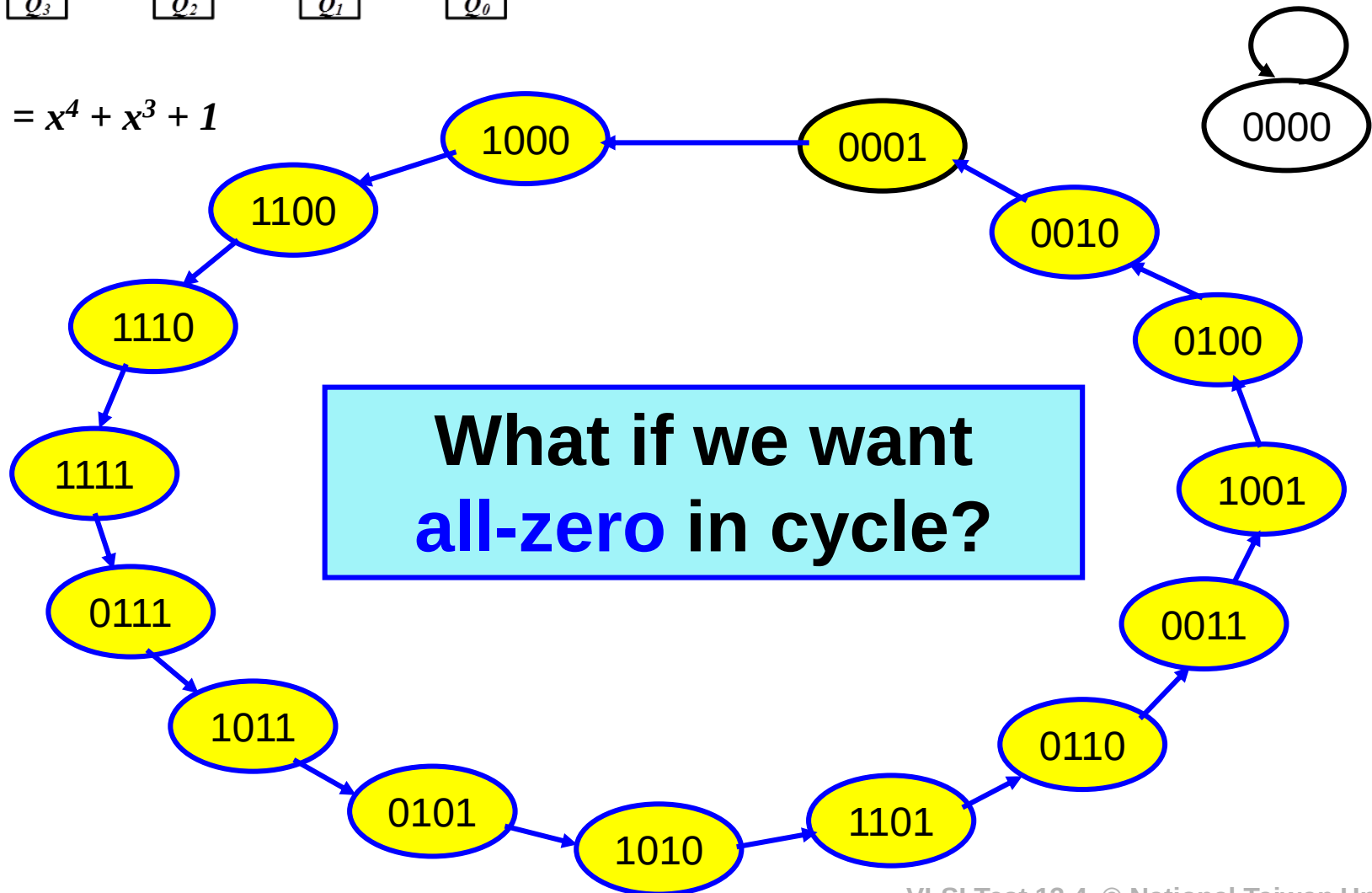
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State Diagram

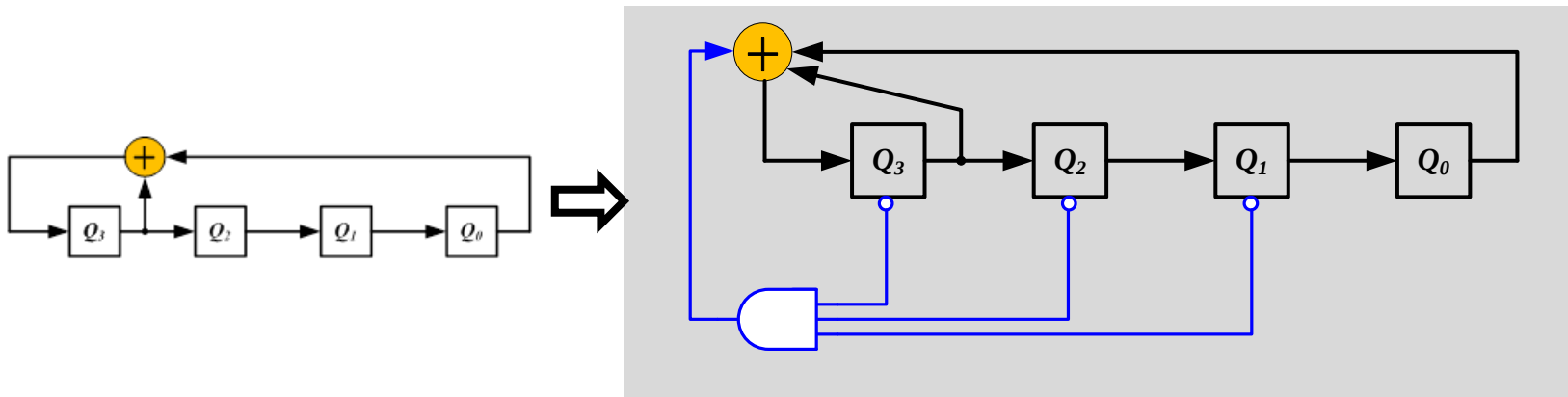


$$f(x) = x^4 + x^3 + 1$$



Exhaustive Feedback Shift Register

- Exhaustive feedback Shift Register (aka. **De Bruijn Counter**)
 - ♦ 2^N different states
- Example: $N=4$; **16** exhaustive states
 - ♦ $0001 \rightarrow \underline{0000} \rightarrow 1000$

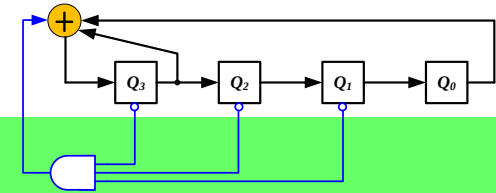
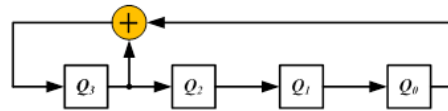


- NOTE: This is NOT LFSR because AND is **nonlinear**

DB Counter Generates All-zero Patterns

Quiz

Q: Show this DB counter generates all 16 states.
Fill in table.

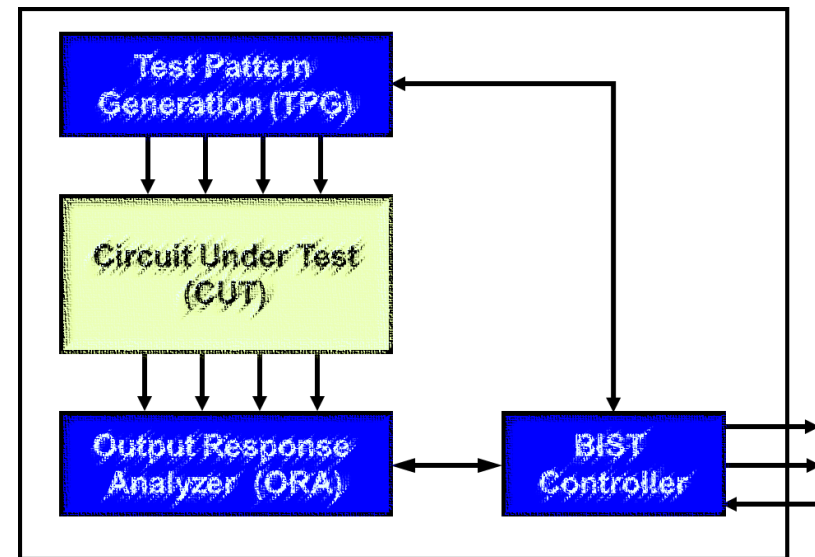


state	Q ₃	Q ₂	Q ₁	Q ₀
0	1	0	0	0
1	1	1	0	0
2	1	1	1	0
3	1	1	1	1
4	0	1	1	1
5	1	0	1	1
6	0	1	0	1
7	1	0	1	0
8	1	1	0	1
9	0	1	1	0
10	0	0	1	1
11	1	0	0	1
12	0	1	0	0
13	0	0	1	0
14	0	0	0	1
15 =0	1	0	0	0

state	Q ₃	Q ₂	Q ₁	Q ₀
0	1	0	0	0
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				

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 - ⇒ What is LFSR degree?
 - * Cellular Automata, CA (1984)



What is LFSR Degree?

[Konemann 91] *not in exam

- Suppose N -degree LFSR, S care bits (i.e. specified bits)
 - ♦ N variables, S equations 
 - ♦ What is minimum N to guarantee a solution (seed) can be found?
- Probability of not finding a solution

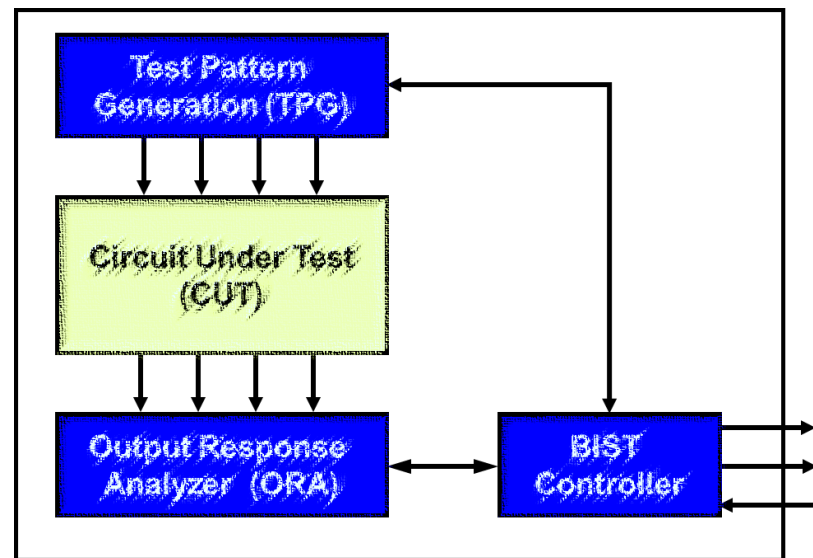
$$P(S, N) = \prod_{i=0}^{S-1} \frac{(2^N - 1) - (2^i - 1)}{(2^N - 1) - 1} \approx 1 - e^{(-2^{(S-N)})}$$


- ♦ When $N \geq (S+20)$, probability not finding solution is 10^{-6}
- Example: scan chain length = 1,000. Assume 5% care bits
 - ♦ $S = 1000 \times 5\% = 50$; $N = 50 + 20 = 70$
 - ♦ Primitive polynomial: $f(x) = x^{70} + x^{16} + x^{15} + x + 1$

LFSR Degree $N \geq (S + 20)$

Summary

- What polynomial to use?
 - ♦ **Primitive polynomials** generates m-sequence
 - * $L_c = 2^N - 1$
 - * Almost half one and half zero
 - * **Cyclic-shift-and-add** property
- How to generate all-zero pattern?
 - ♦ **Exhaustive feedback shift register**
- How many degree N ?
 - ♦ $N \geq \text{care-bits} + 20$



Pros and Cons of LFSR

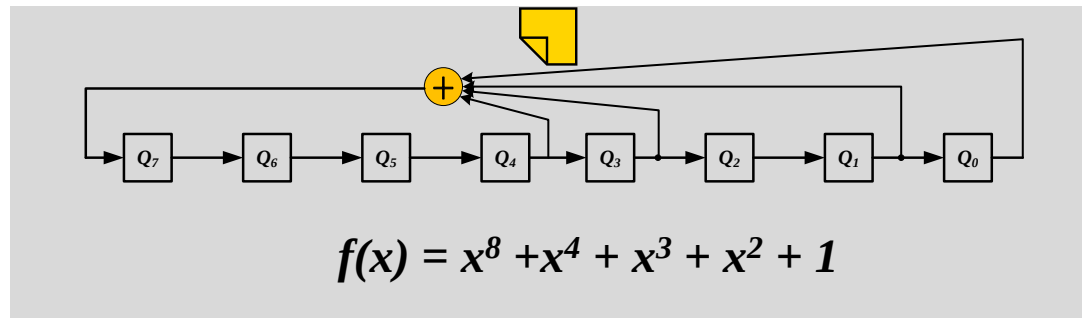
- Advantages

- 😊 Small area.
- 😊 Easy to design: PP → structure
- 😊 m-sequence has nice properties

- Disadvantages

- 😞 Not enough randomness
- 😞 Large LFSR has many tap points. too slow!

How to solve? see Cellular Automata



state	Q ₀	Q ₁	Q ₂	Q ₃
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
4	1	1	0	0
5	0	1	1	0
6	0	0	1	1
7	1	1	0	1
8	1	0	1	0
9	0	1	0	1
10	1	1	1	0
11	0	1	1	1
12	1	1	1	1
13	1	0	1	1
14	1	0	0	1

↔ ↔
1 1

LFSR Is Most Popular TPG

FFT

- Q: *De Bruijn Counter* generates 2^N sequence
 - ♦ $N=4, L_c=16$. 0001111010110010
 - ♦ Does it satisfy cyclic-shift-and-add property?

$$\begin{array}{r} \oplus \quad \begin{array}{|c|c|c|c|c|c|c|c|c|c|c|c|c|c|c|c|} \hline 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \\ \hline \end{array} \\ \hline \\ ? \end{array}$$

